

LLM for Teens

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■ What is an LLM

An LLM is a tool that predicts the next word.

Why LLMs matter

- Because it has seen millions of examples, an LLM can use next-word prediction to do things

like:

- answer questions
- write stories
- summarize long texts
- translate languages
- generate code
- chat naturally
- All of these are just “next word” tasks done extremely well.

LLM vs NLP

- NLP = small tools + rules
- LLM = huge model + deep patterns + transformer architecture
- LLMs are the “super-powered” version of everything teens learned in NLP.

LLM Structure

- An LLM has four main parts.
- Tokens
 - Text is split into small pieces.
- Embeddings
 - Tokens turn into numbers (smart encoding, like your NLP series but contextual).
- Transformer blocks
 - attention
 - feed-forward
 - residual
 - layer norm
- Predicts the next token.

Attention idea

- The model looks at the important words in the sentence.

Attention idea

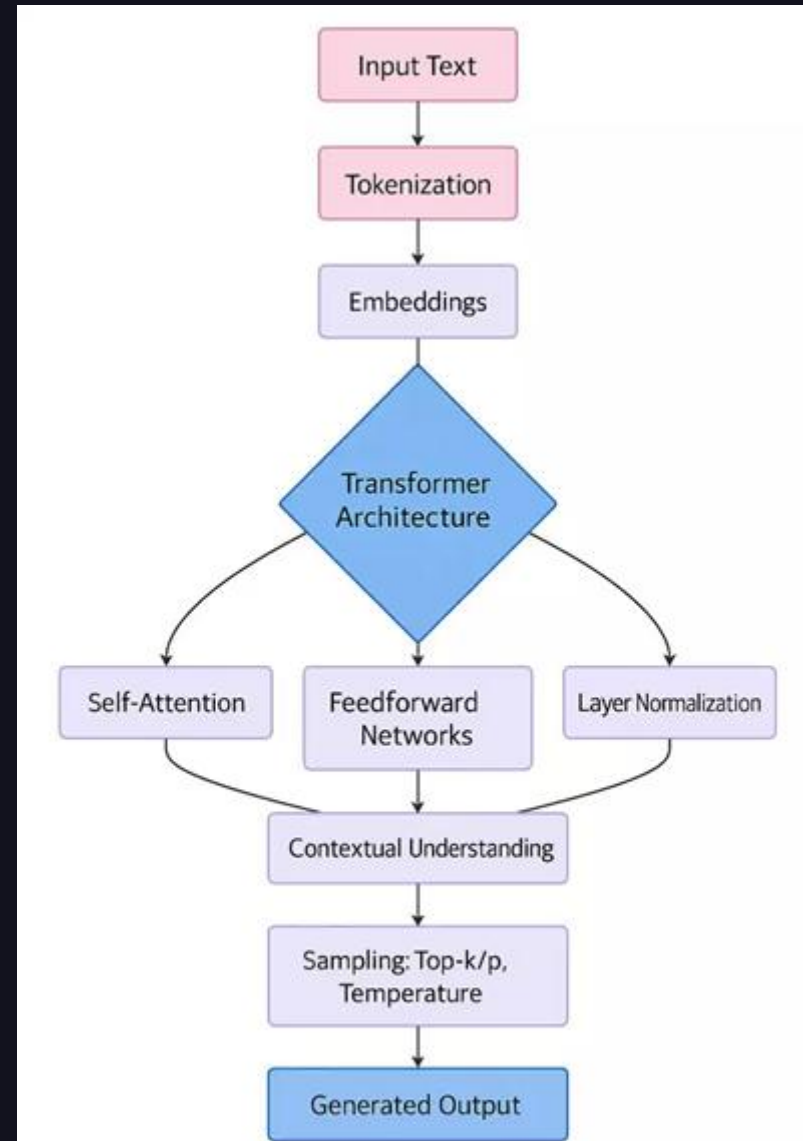


Figure 2: LLM Architecture

Temperature, Top-k , Top-p

- Temperature — adjusts how predictable or creative the model is when choosing the next word.
- Top-k — limits the model to the k highest-probability next words.
- Top-p — limits the model to the most likely words whose combined probability reaches p .

Transformer overview

- A Transformer is a model that uses attention to understand which words matter, and then uses many layers to read or write text.
1. Encoder-only → understanding
BERT, RoBERTa, DistilBERT
Used for: classification, NER, sentiment, embeddings
 2. Decoder-only → text generation
GPT, LLaMA
Used for: chatbots, writing, code
 3. Encoder-Decoder → translation, summarization
T5, BART
Used for: seq-to-seq tasks

■ Intro to RAG

- LLMs are smart, but they don't know your documents, and sometimes they guess wrong. RAG helps by giving the model the correct information from your data so it can answer properly.
- RAG = give the model the right info when needed.
- Fine-tuning = teach the model new behavior.
- or
- RAG = add knowledge.
- Fine-tuning = change the brain.

Assignment

- In this assignment, you will build a small Retrieval-Augmented Generation (RAG) agent. The goal is to demonstrate how an AI model uses your dataset, not only its own training knowledge.
- Dataset
 - You must use the following dataset: Kaggle Bank Customer Complaints

Summary

- ✓ What is an LLM
- ✓ Why LLMs matter
- ✓ LLM vs NLP
- ✓ LLM Structure
- ✓ Attention
- ✓ Transformer overview
- ✓ Temperature / Top-k / Top-p
- ✓ Intro to RAG

Thank You

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